2.8 Squaring and Subtracting Two

WORKSHEET

The question

STEP III 2021, Question 8

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Why can a number in $[-2, 2]$ never escape? What is the largest and smallest that $x^2 - 2$ can be when $-2 \leq x \leq 2$?

STAGE 1 Escaping

Ask yourself Part (i) has a 4^{n-1} in it, suggesting that the distance above 2 at least quadruples at each step. Write $x_n = 2 + d$. What is x_{n+1} in terms of d?

STAGE 2 Escape if and only if $|a| > 2$

Ask yourself "If and only if" means two separate proofs. Which direction does part (i) almost give you, and what is missing?

STAGE 3 A hidden formula

Ask yourself If $a > 2$, there is a number $t > 1$ with $a = t + \dfrac{1}{t}$. (Why?) What are x_2 and x_3 in terms of t?

STAGE 4 The product telescopes

Ask yourself What is $\left(t - \frac{1}{t}\right)\left(t + \frac{1}{t}\right)$? What happens if you multiply the result by $t^2 + \frac{1}{t^2}$?

Pause and try Compute $x_{n+1}^2 - 4$ in terms of t, and compare with the product.

What's really going on

Order and chaos

What happens inside $[-2,2]$? There, $a = t + 1/t$ fails (it needs $|a| \geq 2$), but the trigonometric version works: write $a = 2\cos\theta$. Then $x_2 = 4\cos^2\theta - 2 = 2\cos 2\theta$, and in general

$$x_n = 2\cos(2^{n-1}\theta).$$

So inside the interval the map simply *doubles an angle*. That sounds harmless, but doubling an angle is chaotic. Write θ/π in binary: doubling shifts the binary point one place to the right, throwing away the leading digit. After 50 steps, x_n depends on the 51st binary digit of θ/π, so an error in the starting value of 10^{-15} (the precision of a typical computer) makes the answer after 50 steps completely unreliable. This sensitive dependence on initial conditions is the hallmark of chaos, illustrated in Figure 2.8.1. The map $x \mapsto x^2 - 2$ is equivalent, by a change of variable, to the famous logistic map $x \mapsto 4x(1-x)$, which we meet again in Section 8.7.

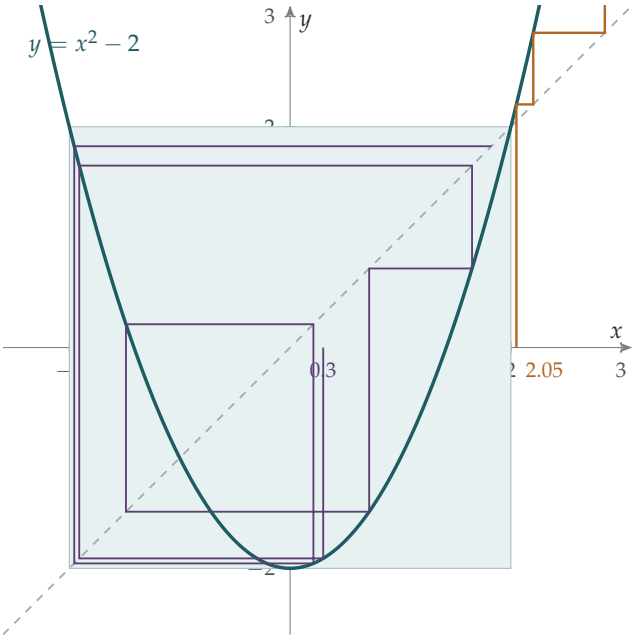


Figure 2.8.1 Cobweb diagrams for $x_{n+1} = x_n^2 - 2$. Starting at 0.3 (plum), the orbit stays in the shaded square $[-2,2]^2$ for ever, wandering chaotically. Starting just outside, at 2.05 (amber), it escapes: $2.05, 2.20, 2.85, 6.13, \ldots$

Finding the largest primes

A *Mersenne number* is $M_p = 2^p - 1$. In 1876 Édouard Lucas found, and in 1930 Derrick Lehmer completed, a remarkable test for whether M_p is prime (for p an odd prime):

start with $s_0 = 4$ and repeatedly square and subtract 2, reducing modulo M_p each time. Then

$$M_p \text{ is prime} \iff s_{p-2} \equiv 0 \pmod{M_p}.$$

For $M_7 = 127$: $s = 4, 14, 194 \equiv 67, 67^2 - 2 \equiv 42, 42^2 - 2 \equiv 111, 111^2 - 2 \equiv 0$, so 127 is prime. The connection with this question is the substitution: $4 = t + 1/t$ with $t = 2 + \sqrt{3}$, so $s_k = (2+\sqrt{3})^{2^k} + (2-\sqrt{3})^{2^k}$, and the proof of the test uses arithmetic with numbers of the form $a + b\sqrt{3}$, rather like the Gaussian integers of Section 1.6. Because each step needs only one squaring, the test is extremely fast, which is why most of the very largest known primes, including the current record holder, are Mersenne primes (not all of them are: other special forms, such as $a^{2^n} + 1$, have their own fast tests). As of the time of writing the record is $2^{136279841} - 1$, a number with more than 41 million digits, found in 2024.

The Mandelbrot set

Replace -2 by any constant c and allow complex numbers: $z \mapsto z^2 + c$. The set of c for which the orbit of 0 stays bounded is the *Mandelbrot set*. Our map has $c = -2$, the leftmost point of the Mandelbrot set, and the set of starting values whose orbits stay bounded (its *Julia set*) is exactly the interval $[-2, 2]$ that we found in part (ii). For almost every other c, that set is a fractal.

Try this next

1 **Inside the interval.** Prove that if $a = 2\cos\theta$ then $x_n = 2\cos(2^{n-1}\theta)$. Find all values of a for which $x_3 = a$ but $x_2 \neq a$. (*Hint:* you want $4\theta \equiv \pm\theta$, modulo 2π.)

2 **Lucas–Lehmer by hand.** Use the test to show that $M_5 = 31$ is prime and that $M_{11} = 2047$ is not. (For M_{11} you will need nine steps; a calculator helps. Check your conclusion by factorising 2047.)

3 **An explicit example.** Take $A = \frac{3}{2}$. Find a and t, and check the formula $y_1 = \sqrt{x_2^2 - 4/x_2}$ numerically.

4 **A sum that telescopes.** HARDER With $a = t + 1/t$ and $t > 1$, show that

$$\frac{1}{x_1 x_2 \cdots x_n} = \frac{t - t^{-1}}{t^{2^n} - t^{-2^n}}.$$

Using the identity $\dfrac{1}{z-1} - \dfrac{1}{z^2-1} = \dfrac{z}{z^2-1}$ with $z = t^{2^n}$, deduce that

$$\frac{1}{x_1} + \frac{1}{x_1 x_2} + \frac{1}{x_1 x_2 x_3} + \cdots = \frac{1}{t}.$$

Check this numerically when $a = 3$.

Further reading

- **Ian Stewart, *Does God Play Dice?*** (Penguin). The classic popular account of chaos, including the logistic map.
- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). The standard undergraduate introduction; Chapter 10 is about one-dimensional maps like this one.
- **Marcus du Sautoy, *The Music of the Primes*** (Fourth Estate). Includes the story of the hunt for Mersenne primes.

The full guided solution is in Section 2.8:
https://beyondstep.keithmcnulty.org/ch02/squaring-minus-two.html